

A SELF-SUPERVISED FRAMEWORK FOR SPACE OBJECT BEHAVIOUR CHARACTERISATION.

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ABSTRACT

Foundation Models, which leverage large neural networks pre-trained on unlabelled data before fine-tuning for specific tasks, are increasingly being applied to specialised domains. Recent examples include ClimaX for climate and Clay for satellite Earth observation, but a Foundation Model for Space Object Behavioural Analysis has not yet been developed. As orbital populations grow, automated methods for characterising space object behaviour are crucial for space safety. Here, we present a self-supervised framework for space object behavioural analysis, representing a first step towards a Foundation Model for SOBA. The backbone of this framework is a Perceiver-Variational Autoencoder (VAE) architecture, pre-trained with self-supervised reconstruction and masked reconstruction on 227,000 light curves from the MMT-9 observatory. The VAE enables anomaly detection, space object motion prediction, and generation of synthetic light curves. We fine-tuned the model for anomaly detection & motion prediction using two independent light curve simulators (CASSANDRA and GRIAL respectively), using CAD models of boxwing, Sentinel-3, SMOS, and Starlink platforms. Our pre-trained model achieved a reconstruction mean squared error of 0.009, identifying potentially anomalous light curves through reconstruction difficulty. After fine-tuning, the model scored 85% and 82% accuracy, with 0.92 and 0.95 ROC AUC scores respectively in both anomaly detection and motion mode prediction (e.g., sun-pointing, spin, tumbling etc.). Analysis of high-confidence anomaly predictions on real data revealed distinct patterns including characteristic object profiles and satellite glinting. The motion mode prediction model successfully differentiated between various movement behaviours such as sun-pointing, spin, and tumbling. Our work demonstrates how self-supervised learning can simultaneously enable anomaly detection, motion prediction, and synthetic data generation from rich representations learned in pre-training. More broadly, our work supports space safety and sustainability through automated monitoring and simulation capabilities.

Keywords Space Object Behavioural Analysis, Space Foundation Model, Self-Supervised Learning, Light curve anomaly detection, Attitude prediction, Space Situational Awareness (SSA), Generative AI

1 INTRODUCTION

The number of objects launched is growing rapidly, with 159 worldwide launches in 2000 versus 2849 in 2024 [1]. This growth underscores the need for methods for automated and efficient monitoring of space objects, which is crucial to societal function and national security e.g., critical communications & position/navigation/timing [2]. In addition to national security, monitoring of Space Objects has substantial economic considerations, e.g., the financial service sector is reliant on precise time synchronisation enabled by satellites [3]. This increase in space resident objects is coupled with significantly larger datasets collected from modern sensors, tracking objects both from ground-based sensors and space-space sensors [4, 5]. Traditional methods for Space Object Behavioural Analysis (SOBA) rely on arduous manual inspection, or numerical methods requiring a large amount of priors. Machine learning and Artificial Intelligence techniques offer promising new possibilities to analyse these growing datasets. For example, Foundation Models, whereby large neural networks are (pre-)trained

to learn general principles of unlabelled data, are emerging as a powerful approach for pattern recognition in large, unstructured datasets. They have demonstrated performance in multiple natural language and text-based tasks, e.g., writing and coding. However, Foundation Models that integrate sensor data with physics-based models, are still in their infancy, as there are less publicly available datasets and input/output data types are less intuitive than natural language. Nevertheless, domain-specific Foundation models are proving effective in specialised use-cases such as ClimaX [6] for climate/weather prediction, and Clay for Earth Observational data [7]. These examples show that the general principles of learning spatiotemporal relationships between datapoints are readily transferable and useful outside of natural language and programming.

However, there is little current exploration on the best neural network architectures and training strategies for FMs for more specialised datasets/applications. To our knowledge, this work represents the first step towards a Foundation Model for Space Object Behavioural Analysis. Whilst our model is pre-trained

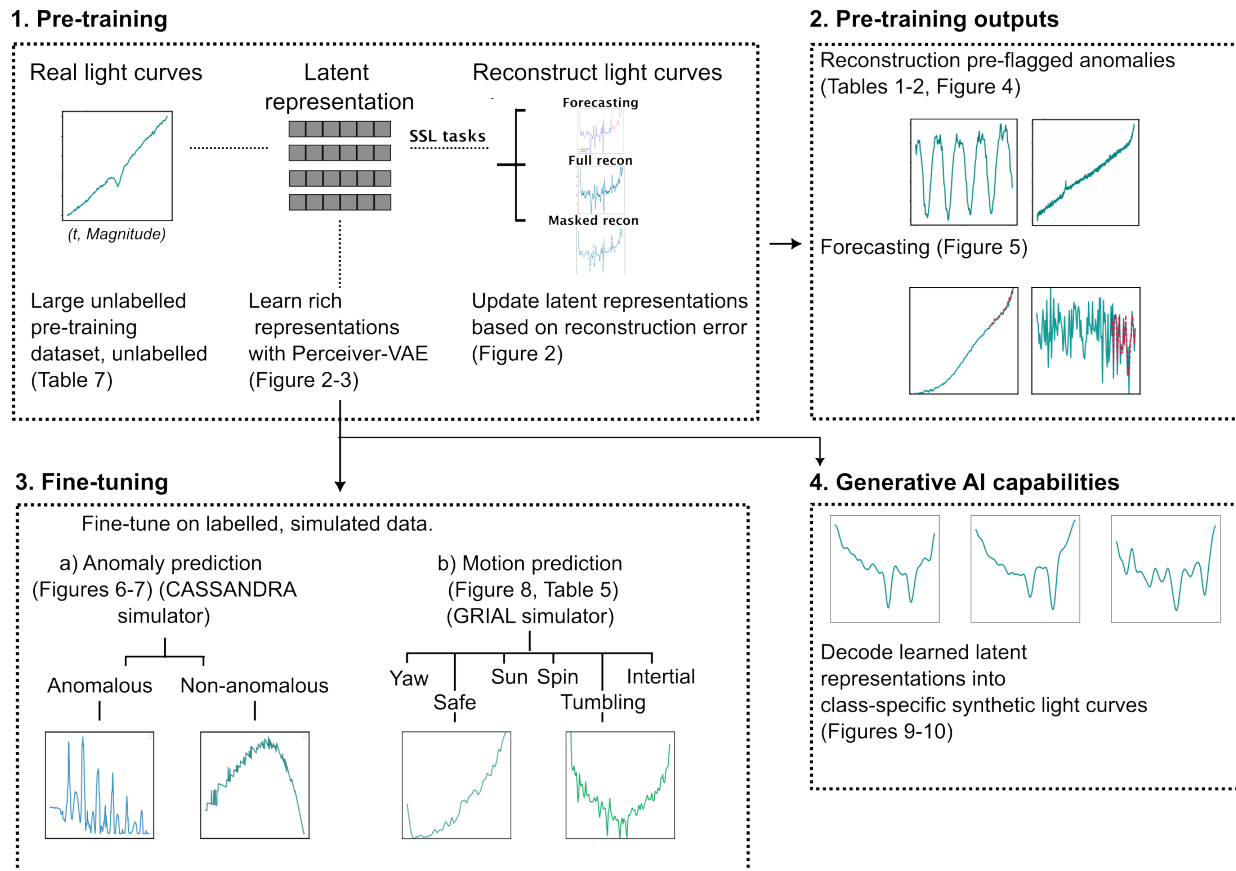


Figure 1: Graphical structure of the paper. First, in Section 2.1 we describe pre-training our model with a large unlabelled dataset of real light curves. We encode these into a rich latent representation with a Perceiver-VAE architecture, updating these representations based on three self supervised learning (SSL) tasks: Reconstruction, Forecasting, and Masking. We next analyse the results of pre-training, which includes pre-flagging of anomalies based on reconstruction difficulty (Sections 2.2-2.3), and forecasting quality of the model (2.4). Following this, we describe fine-tuning our rich representations for two downstream tasks: a) anomaly prediction (Section 2.5, and b) motion prediction (Section 2.6). Finally, we demonstrate further utility of our representations by generating de novo datasets according to a particular motion type (Section 2.7).

on a single modality (light curves), the Perceiver architecture is well suited to multimodal extension, and we discuss this direction in Section 4. We therefore use the term Foundation Model as a framing for this approach, following the precedent of domain-specific models such as ClimaX and Clay. Of principal interest in SDA is anomaly detection, i.e., detecting space objects that are behaving in an unusual way, such as motions, manoeuvres, or those that are performing atypical functions. Characterising these atypical behaviours can be done through simulation of space object observations, but this is challenging, due to imperfect physical models and computational cost. Therefore, general purpose models for SDA should be able to perform anomaly detection, integrating real observations with physical models in a computationally efficient way. Foundation Models are typically pre-trained on large datasets to learn some compressed rich, general latent representation of the data distribution. They are then fine-tuned for specific tasks by further using those latent representations, for example, classifying them according to a small amount of labelled data. Whilst in pre-training, the goal is usually to keep the optimisation task relatively unconstrained to encourage general features to be learned, we can still make architectural decisions to build in

anomaly detection from the ground up through the use of Variational Autoencoder (VAE) components.

VAEs are generative models which are trained by learning to reconstruct the input data from a sampling of the learned latent space [9]. They provide a simple and intuitive way to perform anomaly detection, as they inherently produce a reconstruction error at inference. This error is the difference between the model’s reconstruction of an input and the actual input. If a specific data point is unusual compared to the training data, or is not well represented in the training data, it will produce a high reconstruction error, suggesting anomalous behaviour or properties. Unsupervised models with Variational Autoencoders (VAEs) have been used extensively as anomaly detectors, (for a recent review in the context of solar images see the Introduction in [10]).

Whilst VAEs offer an effective anomaly detection mechanism, they still need integration into a broader neural network framework to effectively process complex, high dimensional space object data. Transformer architectures are an attractive choice here, due to their ability to capture long range dependencies in sequential data. When choosing a Transformer architecture,

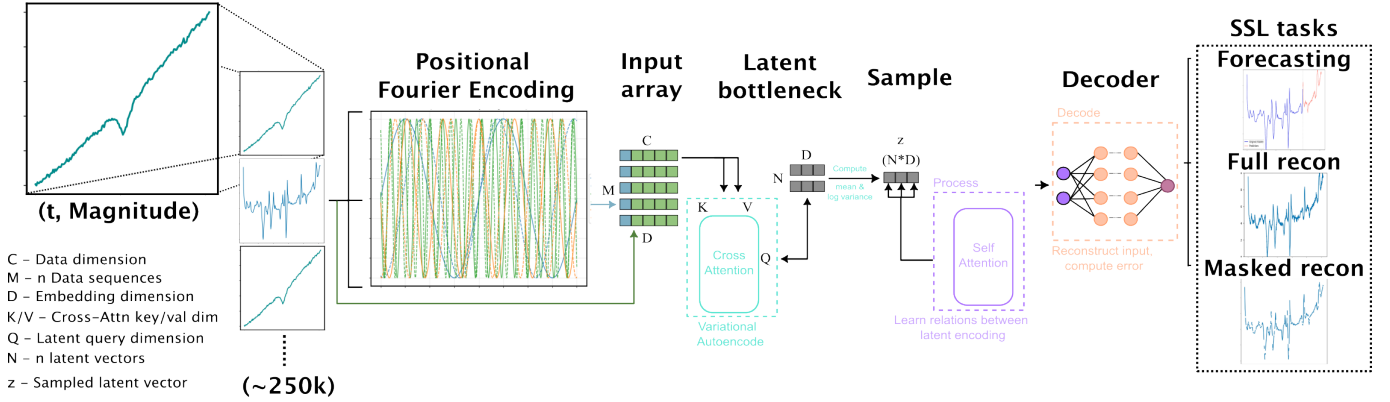


Figure 2: Pre-training approach. Input array(s) (M) provide keys (K) which index the data (e.g., timestep in a timecourse) and values (V) which represent the information at each K . The model includes a learned latent array (N) which provides queries (Q) for the Cross Attention mechanism. Q interacts with K and V to extract relevant information from the input. The latent representation (z) is computed from this mechanism using the mean and log variance, capturing compressed features of the input. In this way, z acts as a latent bottleneck. Self Attention layers then learn meaningful relations within this latent space. This architecture incorporates elements of Variational Autoencoders (VAE) in its training process, whereby the loss is calculated by sampling probabilistically from the latent space. Figure adapted from [8].

there are numerous variants with respective strengths and weaknesses. For example, conventional transformer-based neural networks perform the computationally costly full self attention mechanism between all the inputs. This means that for very long sequences, standard transformers incur quadratic computational cost with respect to sequence length, which can become prohibitive at scale. To address this limitation, DeepMind recently developed the Perceiver/Perceiver-IO, a transformer-based neural network architecture. [8, 11]. Whilst a light-curve specific model may benefit from a state-of-the-art architecture for time-series analysis such as TimesNet [12] for anomaly detection or TimeXer [13] for forecasting, for a Space Domain Awareness Foundation Model we prioritise extensibility to other observational modalities (e.g., hyperspectral curves, ISAR images, attitude measurements, orbital parameters). To achieve this multimodal compatibility, we adopt the Perceiver architecture, which is straightforwardly extendable to multiple modalities through its cross-attention latent bottleneck. This choice also reduces the computational complexity of the vanilla transformer $O(N^2)$ to $O(N)$ with respect to sequence length, an advantage that is primarily forward-looking in this work but could support edge deployment (e.g., on-board a satellite).

Given the potential of VAEs as anomaly detectors, and the computational efficiency of Perceiver, a VAE-based Perceiver model is a promising approach for a computationally efficient, SDA FM for anomaly detection. In this work, we use visible light curves as a test-bed for our self-supervised framework, providing proof of principle for a SDA FM. Light curves are readily available, with large amounts of real observational data, and the maturity of light curve simulators. Here, we use the largest publicly released dataset of light curves, from the Mini-Mega TORTORA (MMT-9) observatory in Russia [14].

Light curves are plotted time-series measurements of the brightness of light received by a sensor reflected by, or emitted from a space object (SO). Light curves allow both the physical properties and the behaviour of the SO to be inferred. For

example, rotating objects may exhibit short-term periodic variations in their light curves.

Light curves are well studied in the literature and recently there has been substantial interest in the field in using light curves to train machine learning models [15, 16, 17]. These efforts are mostly concerned with supervised classification problems; whereby light curves are labelled into classes, and the model is trained to distinguish between those classes. Despite this focus, some groups have trained unsupervised deep learning models for light curve analysis, [18, 19, 20, 21]. However, to our knowledge, our work here is the first to leverage unsupervised learned representations of light curves of SOs from a large dataset to downstream applications, i.e., anomaly detection, object characterisation, and synthetic data generation. Figure 1 outlines the research in this study (into pre-training, fine-tuning, and generative AI capabilities). This paper is structured as follows: Section 2 presents the results of our pre-training approach, anomaly detection, motion prediction fine-tuning, and synthetic data generation capabilities; Section 3 details our methodological approach and implementation. Finally, Section 4 discusses future directions and implications.

2 RESULTS & DISCUSSION

2.1 Pre-training VAE-Perceiver to reconstruct MMT light curves.

First, we trained a VAE-based Perceiver on the MMT-9 light curve dataset. Training showed expected convergence behaviour, with rapid initial improvement followed by gradual refinement. The loss values reduce consistently before plateauing at 0.0011 (train) and 0.0012 (validation) (Figure 3, Left-hand panels). Additionally, when training a VAE, we optimise the latent to space to be well ordered using a KL divergence term (see equation 7). This term measures how normally distributed the latent space is, and therefore can be interpreted as a measure of dataset heterogeneity. The KL divergence terms exhibit interesting dynamics - they increase early in training, peaking at ap-

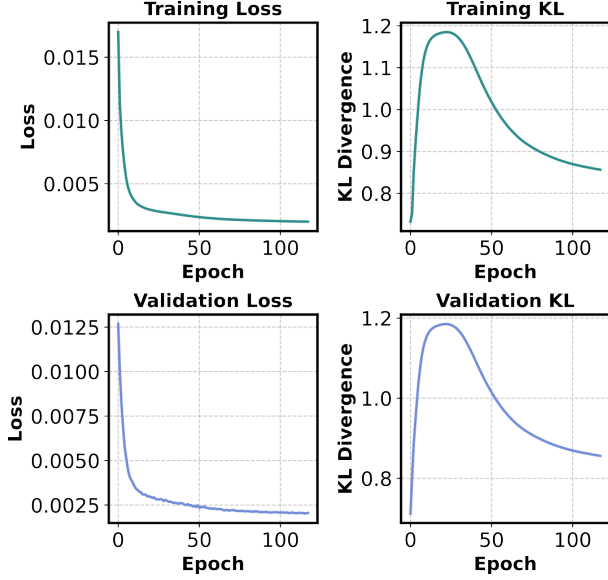


Figure 3: Training curves for a VAE-Perceiver model trained on the MMT-9 light curve dataset. Training and Validation losses decrease from >approximately 0.015 to 0.0011 and 0.0012 respectively. The KL Divergence of the latent distribution initially increases, before plateauing at approximately 0.85. Validation loss decreases to a similar plateau, without the initial first epoch decrease.

proximately epoch 25, then gradually reduce and stabilise. This suggests the VAE initially encodes maximum information into the latent space, before finding a balance between reconstruction accuracy and latent space regularity. The close tracking of train and validation loss suggest a good generalisation performance, without memorisation. Final KL Divergence values of approximately 0.86 indicate the model maintained a meaningfully ordered latent space whilst avoiding latent space collapse (recall that a KL divergence of 0 implies a normally distributed latent space, Figure 3, right-hand panels). As a VAE loss function minimises the discrepancy between reconstruction and input data, the loss values can be interpreted directly as an error score. For our pre-training, the average validation error score was approximately 0.012%.

2.2 Flagging of anomalous Space Objects in the MMT-9 Dataset during pre-training.

Using a held-out test portion of the MMT dataset, we examined the highest and lowest reconstruction error samples (Tables 1-2). Among the highest reconstruction error test samples (Table 1), we found a mixture of satellites and rocket bodies, some appearing frequently in the pre-training data. For example, light curves from EGS (AJISAI) occur 1064 times in the MMT dataset, yet four light curves appear in the ten highest reconstruction error. Similarly, light curves from GLOBSTAR M092 and CZ-3 R/B (occurring 285 and 269 times respectively) also showed high reconstruction errors. When an object such as AJISAI appears frequently in the pre-training data, the model learns a stable distribution of typical light curve morphologies for that object. A curve that falls outside this learned distribution is therefore more likely to represent atypical behaviour for

that specific object, even if the object itself is well represented. For less frequently observed objects, the model has less prior information and cannot make the same distinction. This motivates the use of large, diverse pre-training datasets to maximise the range of objects for which such distinctions can be made. For the lowest reconstruction error samples (Table 2), we again find a mixture of satellites and rocket bodies, with one example of debris, typically with fewer appearances in the pre-training data (ranging from 9–153).

2.3 Analysis of pre-training early-flagged Space Objects

To understand the nature of these highest/lowest reconstruction error curves, we manually inspected them (as shown in Figure 4). The ten highest reconstruction error curves were all highly periodic (Figure 4 A-J), with frequent large changes in maxima and minima. In contrast, the lowest error curves were substantially more linear (Figure 4 K-T). Whilst most were monotonic, panels K, P, T showed small deviations from this pattern.

Highly periodic curves typically represent rapid tumbling behaviour of the space object as highlighted by [17]. Conversely, linear curves indicate objects with smaller attitude changes or highly regular morphology. This analysis supports our presumption that tumbling behaviours in the highest reconstruction errors likely represent atypical SO behaviour. Thus, frequent appearances in training data help the model flag what constitutes unusual behaviour for particular objects.

2.4 Light curve forecasting

An additional output from pre-training is forecasting future lightcurve states from past states. By including forecasting as a self supervised task (see Section 3.3 for details), our model learns a latent space corresponding to different timecourse portions, enabling inference time predictions without fine-tuning. Analysing the same held out test dataset used in Section 2.2, we computed the MSE forecasting loss after masking out the final 25% of the timecourse (see Equation 6). Across approximately 20,000 curves, the mean forecasting error was $7.7e^{-4}$, with a standard error of $7.6e^{-4}$.

To examine the difference in a high/low forecasting error, we visualised the three lowest and highest errors (Figure 5 top and bottom rows respectively). In both cases, the general trends of the light curve were well forecasted (Figure 5 red-dotted line). For the lowest error samples, our model accurately predicted future values with minimal deviation from ground truth. For the highest error samples (predominantly high-frequency periodic curves), the model captured patterns well but struggles with exact magnitude prediction of these features (Figure 5).

This forecasting ability demonstrates that our model has learned meaningful temporal relationships in light curve dynamics unsupervised. Across all the self-supervised learning (SSL) tasks, the reconstruction errors suggest our model has learned rich representations decodable to various light curve types and can potentially identify anomalies. However, making definitive claims remains challenging with unlabelled data, as

Table 1: The ten highest reconstruction errors from the pre-flagged light curve samples contained in the test set. We evaluated the lowest-loss light curve model from Section 2.2 against the entire test dataset unseen during training/validation, recording the reconstruction error for each sample. Examination of the ten highest reconstruction error light curves shows a variety of space objects (SOs). Seven light curves are from satellites (all of which are in orbit), and three belong to rocket bodies. US: United States of America, IND: India, PRC: China.

Intl Code	NORAD	Name	Source	App.	Launch Date	Status	Notes
2019-029N	41912	CZ-3B R/B	PRC	45	2017-01-05	Not in orbit	Rocket Body
1996-017A	28646	USA 182	USA	66	2005-04-30	Unknown	Satellite (classified)
1965-016J	16908	EGS (AJISAI)	JPN	1064	1986-08-12	In orbit	Geodetic
1998-026B	16908	EGS (AJISAI)	JPN	1064	1986-08-12	In orbit	Geodetic
2013-046E	44796	CZ-3B R/B	PRC	138	2019-11-23	In orbit	Rocket Body
2011-061A	38043	GLOBSTAR M092	Globalstar	285	2011-12-28	In orbit	Cellular
2019-082B	16908	EGS (AJISAI)	JPN	1064	1986-08-12	In orbit	Geodetic
1963-038A	29517	CZ-3 R/B	PRC	269	2006-10-28	In orbit	Rocket Body
2012-056D	16908	EGS (AJISAI)	JPN	1064	1986-08-12	In orbit	Geodetic
2013-055G	41857	YUNHAI 1	PRC	48	2016-11-11	In orbit	Polar weather

Table 2: The ten lowest reconstruction error light curve samples from the test dataset. We evaluated the lowest-loss model from Section 2.2 against the entire test dataset unseen during training/validation, recording the reconstruction error for each sample. Examination of the ten lowest reconstruction error light curves shows a variety of space objects (SOs). Seven light curves are from satellites (of which one is no longer in orbit), two belong to rocket bodies, and one to debris. JPN: Japan, US: United States of America, IND: India, PRC: China.

Intl Code	NORAD	Name	Source	App.	Launch Date	Status*	Notes
2006-002A	28931	ALOS	JPN	67	2006-01-24	In Orbit	Remote Sensing
2019-029N	44247	STARLINK-33	Starlink	11	2019-05-24	Not in Orbit	Cellular
2019-082B	44820	CZ-4C R/B	PRC	9	2019-11-27	In Orbit	Rocket Body
1999-057A	25940	CBERS 1	PRC / BR	81	1999-10-14	In Orbit	Earth Resources
2019-074AD	44740	STARLINK 1035	Starlink	28	2019-11-11	In Orbit	Cellular Data
2010-009E	36417	CZ-4C DEB	PRC	153	2010-03-05	In Orbit	Debris
2010-059A	37214	FENGYUN 3B	PRC	133	2010-11-04	In Orbit	Weather
1996-017A	23827	IRS B3	IND	87	1996-03-21	In Orbit	Remote Sensing
2009-066B	36105	H-2A R/B	JPN	24	2009-11-28	In Orbit	Rocket Body
2013-055G	39271	CUSAT 2/FALCON	US	74	2013-09-29	In Orbit	Amateur Radio

the exact events/behaviours causing normal/anomalous patterns are unclear. While step changes might represent manoeuvres or morphological features, validating these hypotheses requires supervised learning or physical modelling. Therefore, we next examined fine-tuning with a synthetic, labelled dataset.

2.5 Fine-tuning for anomaly detection

Obtaining real observational data of satellites with known anomalies is extremely challenging, as the full set of possible anomalies is not well understood or catalogued, and organisations are opaque about anomalous space object behaviour for national security/commercial reasons. To address this data gap, we produced a synthetic dataset for fine-tuning using the framework CASSANDRA (Computational Agent For Space Situational Awareness And Debris Remediation Automation). Within this framework is an orbital simulator first introduced by Vasile et al. (2023), generating the light curves using a CAD model of an orbiting Space Object with a specific geometry. This dataset contains 800 simulations of four different satellite platforms: a boxwing satellite (e.g., Jason-3), Sentinel-3, SMOS, and Starlink. These space objects (SOs) orbit an

observing ground station, with both light and spectral curves recorded as time courses. Within these 800 simulations, a subset contain anomalous events of varying magnitudes (specifically debris collisions that result in detectable changes in signal). These changes in signal are shown in Figure 6 F, H, where otherwise linear light curves abruptly change to a different profile, oscillating in magnitude between very high and low minima. We first evaluated the anomaly detection capability of the largest pre-trained model without any labelled data, we passed a balanced set of 32 anomalous and 32 normal CASSANDRA light curves through the frozen encoder and used reconstruction error as an anomaly score, flagging the top 10% highest errors as anomalous. This unsupervised approach achieved 61% accuracy and a ROC AUC of 0.71, demonstrating that the pre-trained representations capture anomaly-relevant structure without any supervision. Fine-tuning with labelled data substantially improves on this, as described below. To fine-tune our pre-trained Perceiver-VAE for anomaly detection, we freeze the encoder weights and train a lightweight MLP classification head on the learned latent representations (see Section 3.4 for details).

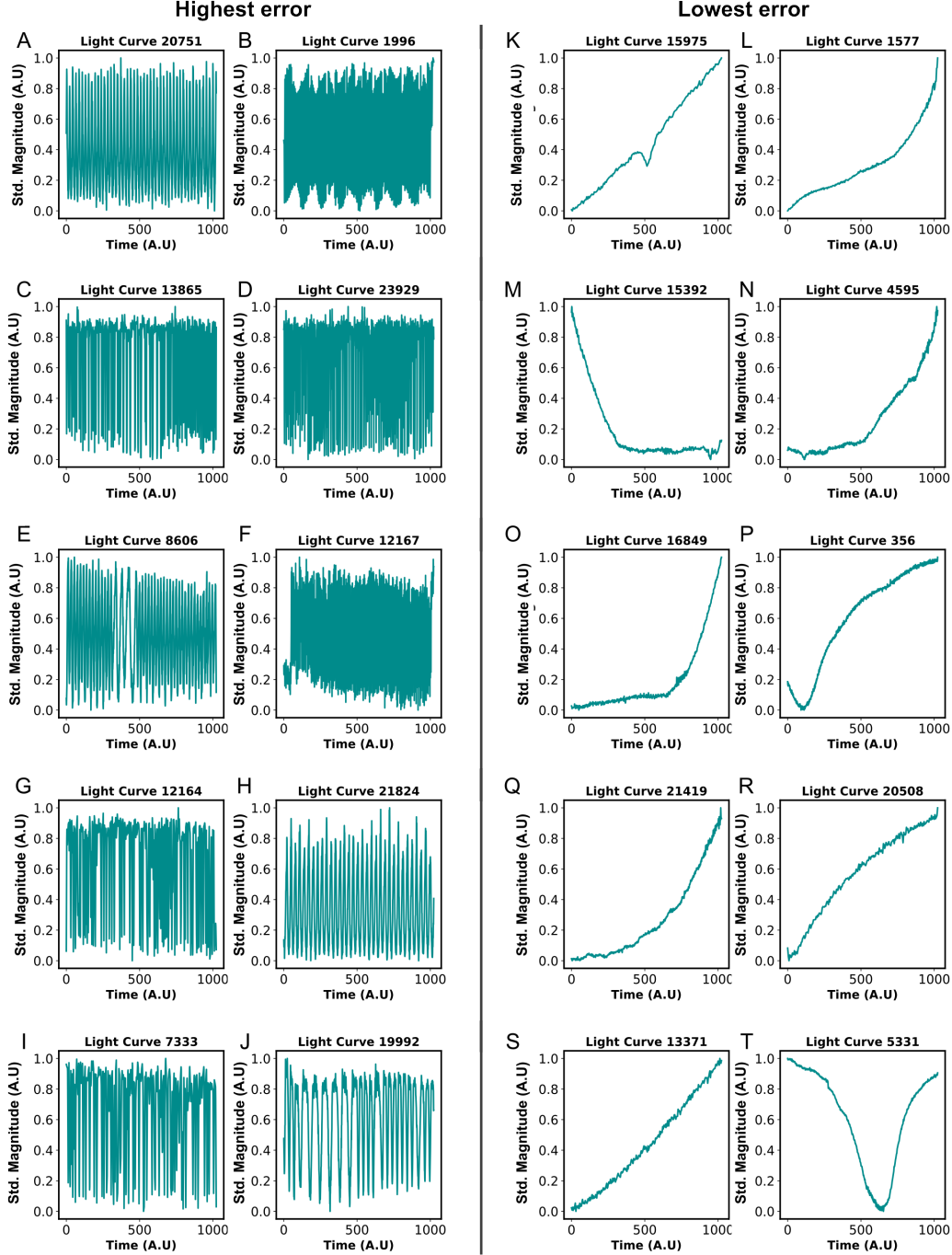


Figure 4: The ten highest reconstruction error test light curves, where A is the highest, and J the tenth highest. These curves exhibit periodic variation over time in the measured standardised magnitude (K-T). As in A-K but for the ten lowest error test light curves, which exhibit mostly linear variation in the standardised magnitude over time. Some curves exhibit troughs in standardised magnitude (K, P, T).

We compare against four supervised baselines trained from scratch on raw light curves, as well as randomly initialised Perceiver encoders (r.e.) at increasing scale (Table 3). Among the baselines, the CNN achieves the strongest ROC AUC (0.849), with the LSTM achieving the best accuracy (0.856). Randomly initialised Perceivers are broadly competitive with these baselines, suggesting the architecture itself has useful inductive bi-

ases for this task. Pre-training consistently improves on random initialisation across all scales, with pre-trained models achieving ROC AUC of 0.911–0.916. Our full model achieves 0.850 accuracy, 0.916 ROC AUC, and 0.793 macro F1, outperforming all baselines across metrics.

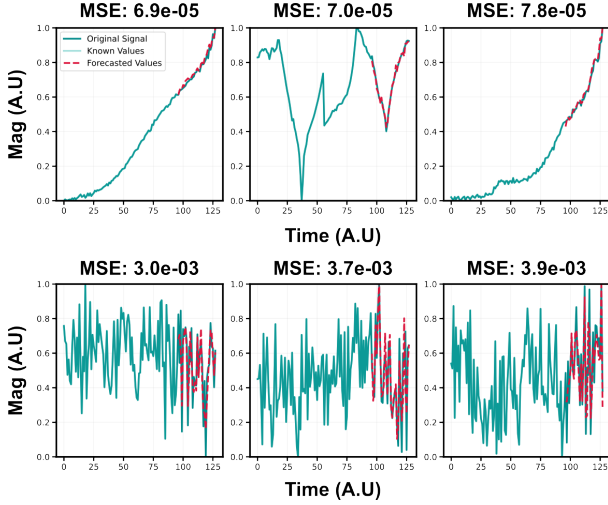


Figure 5: Test light curves with the latter 25% of the timecourse masked at inference. Our Perceiver-VAE was trained with several self-supervised tasks—including a forecasting loss (see Section 3.3 for details). This means that at inference, the future state of a light curve can be predicted by the pre-trained model. **Top row:** the lowest three forecasting mean squared error test samples. **Bottom row:** the highest three forecasting mean squared error test samples. Qualitatively, the original signal (turquoise) and known values (cyan) are aligned. In the lowest error samples, the forecasted values also align well with the original signal, capturing the dynamics well. In the highest error samples, the general trends of the masked regions are successfully forecasted, but the magnitude of the peaks/troughs in the signal is not fully captured.

To assess label efficiency, we evaluated the pre-trained Perceiver against the CNN baseline in a k -shot regime, training each model on $k \in \{2, 5, 10, 25\}$ labelled examples per class across 20 independent runs (Table 4). The pre-trained Perceiver outperforms the CNN at all values of k , including as few as two labelled examples per class (ROC AUC 0.777 ± 0.082 vs. 0.731 ± 0.049). The advantage of pre-training grows with k , reaching a gap of 0.082 AUC points at $k = 25$. This demonstrates that the representations learned during pre-training are particularly valuable in the label-scarce regime typical of anomaly detection in operational SDA settings.

Having fine-tuned our model to high classification accuracy, we next analysed the held-out test dataset, classifying test samples as anomalous/non-anomalous and visualising the top twelve most confidently predicted anomalies (Figure 7). We found a variety of SOs in this set, including several satellites (Figure 7 A, B, E, H, J, L). All of these satellites except the light curve belonging to FLOCK 4P 6 (Figure 7 E) exhibited ‘J’ shaped curves, which simulations suggest may represent glinting from a highly reflective surface aligning with the sun [22]. We also detected light curves belonging to debris (Figure 7 C, D, G), all displaying prominent ‘U’ shaped profiles i.e., where the minima in magnitude is approximately in the middle of the observation. Finally, we identified three other distinct light curve motifs (Figure 7 red dotted outlines): abrupt changes in magnitude which presented as peaks (Figure 7 E), step changes (Figure 7 F), or highly periodic variation (Figure 7 K). We note

Table 3: Anomaly detection performance on the held-out test set. Supervised baselines are trained from scratch on raw light curves. Perceiver models are shown at increasing scale; random encoder (r.e.) variants confirm that pretrained representations drive performance.

Model	Size	Accuracy	ROC AUC	Macro F1
RNN	—	0.831	0.635	0.587
LSTM	—	0.856	0.741	0.783
CNN	—	0.819	0.849	0.743
Transformer	—	0.800	0.774	0.672
Perceiver (r.e.)	s	0.756	0.747	0.559
Perceiver (r.e.)	m	0.756	0.734	0.559
Perceiver (pre.)	s	0.825	0.911	0.763
Perceiver (pre.)	m	0.856	0.912	0.796
Ours (pre.)	l	0.850	0.916	0.793

Table 4: Label efficiency comparison for anomaly detection. ROC AUC (mean $\pm$ std. dev. across 20 independent runs) at increasing numbers of labelled training examples (k per class). The pre-trained Perceiver (large) outperforms the CNN baseline at all values of k .

Model	ROC AUC by k shots per class			
	$k = 2$	$k = 5$	$k = 10$	$k = 25$
CNN (baseline)	0.731 ± 0.049	0.754 ± 0.034	0.754 ± 0.009	0.780 ± 0.008
Perceiver (pre.)	0.777 ± 0.082	0.800 ± 0.109	0.851 ± 0.040	0.862 ± 0.030

that whilst these predictions are made on real MMT-9 observations, no ground truth anomaly labels exist for this dataset and therefore real anomalies cannot be confirmed this way. The identified morphologies are therefore interpreted qualitatively, guided by known physical signatures from the literature and simulation. Systematic validation against catalogued anomaly events or operator records remains an important direction for future work.

2.6 Fine-tuning for motion prediction

We also fine-tuned on a separate dataset with ground-truth motions, simulated by the GRIAL simulator from GMV [23]. Example light curves from this dataset are visualised in Figure 8, here we briefly summarise the dataset.

The GMV dataset is a simulated dataset of 22,009 light curves of Sentinel-3A simulated under 10 different motion laws as seen from 30 different ground stations. These motions can be grouped into three distinct behaviours: First, Sun oriented: **SafeX**, **SafeZ** (where the Space Object’s X/Z axis points to the Sun respectively, and Y initially points to the celestial North pole, and the X/Z rotates respectively), and **Sun** (X-axis points to Sun, Y-axis to Celestial North pole, differentiated from Safe by a varied X-axis phase angle. Next, Earth oriented motions: **YAWXC**, **YAWZC**, **YAWXS**, **YAWZS** (where X/Z indicates the Space Object’s nadir pointing axis, and C/S indicates whether the object’s motion is compensating (C) for optical distortion, or (S) maximising solar array lighting). Finally, the dataset also contains labelled general motions less related to satellite function: Tumbling (uncontrolled/complex rotation of the SO), Spin (rotation around a single axis), Inertial pointing

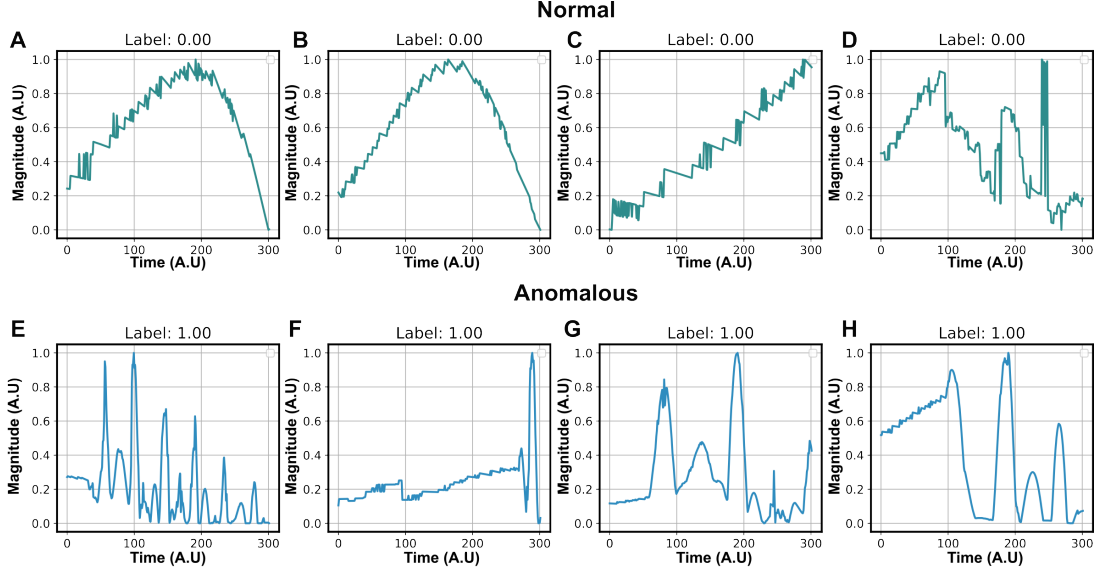


Figure 6: Example normal (A-D) and anomalous (E-H) light curves generated by CASSANDRA as a fine-tuning dataset. Normal curves are typically smooth bell-shaped profiles (A-B), monotonic increasing curves (C), or non-periodic variation with moderate changes in amplitude (D). Anomalous curves contain a variety of distinct characteristic features, such as high-frequency periodicity (E), isolated spikes on otherwise monotonic signals (F), irregular periodic profiles (G), and abrupt extended minima followed by a transition to a periodic profile (H).

Table 5: Comparison of motion classification performance on the held-out test set. Supervised baselines are trained from scratch on raw light curves. Perceiver models are shown at increasing scale; random encoder (r.e.) variants use randomly initialised weights to confirm that pretrained representations drive performance. Our full model (end-to-end finetuning) achieves best performance.

Model	Size	Accuracy	ROC AUC	Macro F1
RNN	–	0.354	0.686	0.273
Transformer	–	0.521	0.753	0.355
LSTM	–	0.693	0.920	0.652
CNN	–	0.711	0.911	0.644
Perceiver (r.e.)	s	0.471	0.625	0.199
Perceiver (r.e.)	m	0.432	0.500	0.101
Perceiver (pre.)	s	0.635	0.860	0.541
Perceiver (pre.)	m	0.638	0.871	0.556
Perceiver (pre.)	l	0.733	0.907	0.628
Ours (pre., e2e)	l	0.827	0.951	0.773

(where the SO has a fixed orientation relative to the J2000 reference frame). To fine-tune our pre-trained Perceiver-VAE, we first group similar motions together, as domain expert manual inspection between e.g., different yaw axial motions is a substantially more challenging task than distinguishing between different motions altogether (e.g., tumbling/sun pointing). After fine-tuning for 100 epochs, our motion classification tuned Perceiver-VAE could classify held out light curves with 82.74% accuracy, with an average ROC-AUC of 0.95 (Table 6). There was inter-class variation in the fine-tuned classifier’s performance. For example, both tumbling and spin show high performance, with precision of approximately 0.97 and 0.90 respec-

tively. Similarly, these classes also showed high recall, both with approximately 0.93. In contrast, inertial has high precision (0.86) but low recall (0.33). This is potentially due to morphological similarity between inertial and yaw light curves (the largest class), both of which exhibit smooth, monotonically increasing profiles without substantial local variation in magnitude (Figure 8, compare A and E). Table 5 compares our model against supervised baselines and randomly initialised Perceiver-VAEs of varying sizes. Among the baselines, the LSTM and CNN models achieve the strongest performance (ROC AUC 0.92 and 0.91 respectively), with both the RNN (ROC AUC 0.69) and Transformer (ROC AUC 0.75) weaker on this task. Similarly, randomly initialised perceiver encoders perform poorly (ROC AUC 0.63 and 0.50 for the small and medium variant respectively). Pre-training provides substantial improvements as feature extraction at all scales, with the pre-trained Perceiver-VAEs reaching 0.86-0.91 ROC AUC. Our full large model uses end-to-end fine-tuning of the pre-trained encoder alongside the classification head (see Section 3.4 for details), which further increases classification metrics (0.95 ROC AUC, with 83% accuracy and 0.77 F1 score).

2.7 Synthetic data generation

Alongside anomaly detection and attitude mode prediction, we can test the utility of our learned embeddings in producing synthetic data.

The generation of useful and representative training datasets for Space Object (SO) behavioural analysis presents us with several challenges. First, real-world observations are limited by various constraints, such as telescope location, atmospheric conditions etc. Because of this, collecting datasets which are

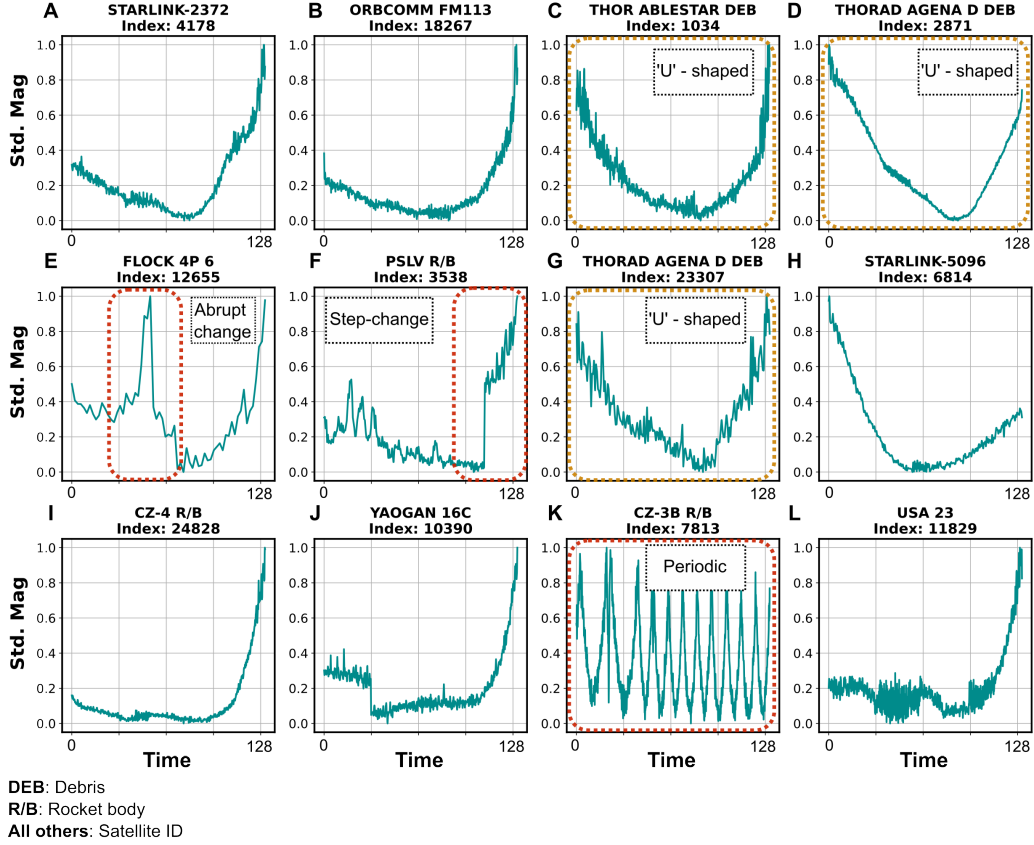


Figure 7: The top 12 confidence anomaly predictions from the fine-tuned light curve Perceiver-VAE model on an independent test set of MMT light curves. We mapped the light curves back to the names of the object, finding a mixture of satellites (A, B, E, H, J, L), debris (C, D, G), and rocket bodies (F, I, K). Within these predicted anomalies, we see notable anomalous patterns such as abrupt peak changes in magnitude (E, red dotted outline), step changes (F, red dotted outline), and highly periodic curves (K, red dotted outline). In addition to well documented anomaly profiles, we also identified characteristic debris profiles. Here every debris light curve has a symmetric 'U' shaped profile, where the minima in magnitude lies approximately in the midpoint of the observation (C, D, G, orange dotted outlines). Units: Std. Mag: Standardised Magnitude rescaled between 0 and 1 (A.U). Time: Resampled to 128 observations (A.U).

Table 6: Performance evaluation of the motion classification model on a held-out test dataset. The model achieves 82.74% overall accuracy with an average ROC AUC of 0.9512. The metrics demonstrate varied performance across different motion classes, with TUMBLING and SPIN showing the strongest classification performance, while INERTIAL exhibits high precision but low recall. For YAW, the most common class, the model shows high recall but lower precision. n_{train} denotes the number of training samples per class.

Motion Class	Precision	Recall	F1-Score	ROC AUC	n	n_{train}
INERTIAL	0.860	0.328	0.475	0.869	262	1,259
SAFE	0.771	0.749	0.760	0.948	342	1,940
SPIN	0.900	0.929	0.915	0.991	225	1,280
SUN	0.676	0.632	0.653	0.945	152	940
TUMBLING	0.970	0.928	0.949	0.997	278	1,333
YAW	0.811	0.972	0.885	0.957	942	5,131
Overall Accuracy	—	0.827	—	0.951	2201	11,883

comprehensive across all the various operational modes, manoeuvres and possible anomalous behaviours requires an unre-

alistic amount of observation time and resource use. Simulators can partially address these challenges but are often con-

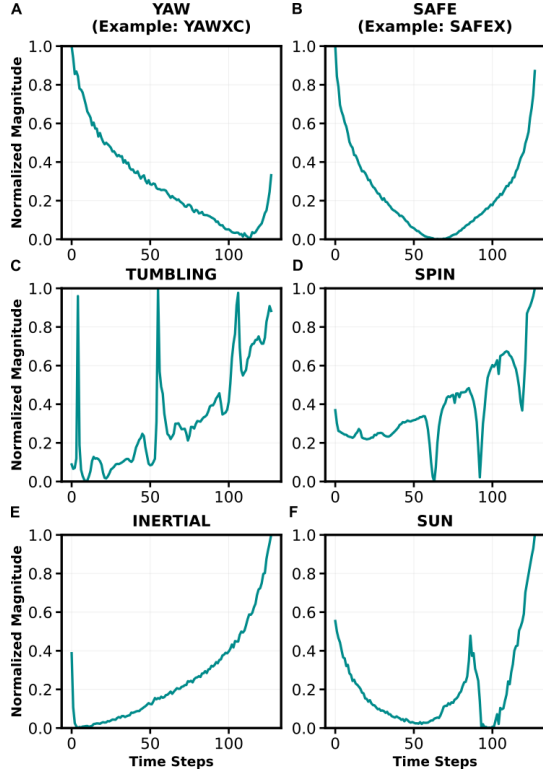


Figure 8: Example light curves from the GMV GRIAL motion finetuning dataset. (A-F) Six grouped motions are labelled into classes. (A) YAW (grouped) YAWXC, YAWXS. The Yaw motions represent the satellite pointing its X axis to nadir, compensating for either optical distortion (XC), or maximising solar array lighting (XS). (B) SAFE (grouped) SAFEX, SAFEZ. The Safe motions represent celestial North pole pointing, with either its' X (SAFEX) or Z axis (SAFEZ) pointing to the Sun. (C) Tumbling, representing uncontrolled/complex rotation. (D) SPIN, representing simpler axial rotation. (E) INERTIAL, fixed pointing, and (F) SUN, representing the space object's X axis pointing to the sun, with Y to the celestial North pole, differentiated from SAFEZ by an X-axis phase angle.

strained by computational costs and the reliability of the underlying physical model.

For these reasons, a method of generating a large amount of diverse physically possible synthetic light curves would substantially enhance training datasets for downstream SDA tasks. Whilst several numerical light curve simulators exist (e.g., we use two distinct software to produce fine-tuning light curves in this study), producing light curves numerically often requires a large amount of priors and a long time to simulate. For example CASSANDRA produces approximately 1000 light curves per hour of simulation. In contrast, producing novel synthetic light curves from the Perceiver-VAE model only requires decoding the learned latent vectors, which is computationally undemanding. This process scales easily, and our initial unoptimised tests presented below generate approximately 40,000 light curves per hour.

If we first recall, that by training a Variational Autoencoder, we are encoding a latent space that is continuous, sampling from

this, then decoding the samples. In addition to enforcing a well-structured latent space, this continuity also implies a functionally infinite number of decodable latent vectors, and therefore also of synthetic data generation.

There are two key considerations when selecting the latent vectors to generate synthetic data. First, physical possibility/plausibility, which ensures generated curves represent possible SO behaviours rather than artifacts of the autoencoding process. Second, the utility of the synthetic data, whereby we want to generate synthetic data that is useful to the field of SDA. To address these concerns, we outline a methodical approach below which emphasises constraining the generation process over sampling in a more general or random way. In other words, potentially the most simple method would be to sample randomly from the learned embedding space, and decode those into light curves. However, whilst this can produce light curves that look roughly realistic, there is no way to validate whether these light curves are physically possible, or to interpret what they represent.

This is a particularly salient concern, as our Perceiver-VAE is pre-trained on many different Space Objects (SOs), including multiple types (e.g., satellites, rocket bodies, debris). It is conceivable that we could index a set of latent vectors which are halfway between encoding a Sentinel-3A light curve and a piece of debris, which is an impossible curve.

Instead, we developed a reference-based sampling approach that ties the synthetic data generation to known physical behaviours whilst allowing for controlled variation. To do this, we queried our embedding space using reference curves as inputs (Figure 9). First, we provided an arbitrary index (Figure 9A), sampling the latent vectors which most strongly activated in response to that input curve, adding varying amounts of noise to those activation vectors, and decoding the resulting vectors into light curves.

Figure 9 shows that we can produce curves with a similar overall trend to the reference curve (i.e., that are appropriately constrained by the reference curve). Secondly, we show that adding increasing amounts of noise to these activations produces curves which vary further from the reference curve (compare Fig 9A, noise scale 1.00 curves to Fig 9A, noise scale 0.25 curves).

Next, we used the fine-tuned model from Section 2.6 to generate motion mode informed synthetic light curves (Figure 9B, C). To do this, we first took a held-out test set used to evaluate the fine-tuned model from Section 2.6. Next we obtained reference curves for the latent space by using light curves with high confidence predictions from that model ($>95\%$ likelihood). Finally, we generated around the neighbourhood of those references by first applying noise to those latent activations and then decoding into light curves as in Figure 9A. Without real labelled light curves to compare against, validating these synthetic data is challenging. To this end, we provide a quantitative assessment of synthetic light curve quality, via validation of the generated curves against a Gaussian Process (GP) fitted to the real light curve distribution for each motion class (Figure 10). For each class, high-confidence real curves (predicted probability >0.75) were used to fit a GP, modelling the class distribution as a non-parametric mean and standard deviation at each time

step. Synthetic curves were then scored by the fraction of time steps falling within the GP’s 95% confidence interval (see Section 3.5 for full details).

Across all six motion classes, synthetic curves scored highly against the GP confidence band, with five of six classes exceeding 94% of time steps within the 95% confidence interval (Figure 10B). The highest scoring classes were TUMBLING (97.6%) and INERTIAL (97.5%), both exhibiting distinctive and consistent morphologies in real data, providing a tight distributional prior for the GP. The lowest scoring class, YAW (89.4%), reflects greater morphological variability within that class — a consequence of YAW grouping four distinct sub-motions (YAWXC, YAWXS, YAWZC, YAWZS), which introduces distributional heterogeneity. These results demonstrate that the synthetic curves produced by our neighbourhood sampling approach are statistically consistent with real light curve distributions across motion classes, without requiring manual smoothing. From this, we saw that in each motion class, we could in essence query an inference dataset to produce more curves of a similar attitude motion mode. Considering that producing these fine-tuning data is time and computationally expensive, we present this and the methodology as something of potentially high utility to the SDA field.

2.8 Model Compression via Magnitude Pruning

To assess parameter efficiency of the large Perceiver-VAE, we applied magnitude pruning separately to the Perceiver layers and the VAE bottleneck. The Perceiver layers tolerate up to 70% weight sparsity with negligible degradation in motion classification accuracy (82.5% vs. 83.0% baseline). In contrast, the VAE bottleneck is considerably more sensitive: 50% sparsity reduces accuracy to 77.2%, a drop of approximately 6 percentage points. This asymmetry suggests that whilst the Perceiver layers contain substantial redundancy at this scale, the bottleneck representations are dense and critical to downstream performance. Re-pretraining a compact architecture informed by these sparsity patterns is a natural next step, discussed further in Section 4. Furthermore, post-training quantisation of the large model to INT8 precision results in negligible loss in motion classification accuracy, whilst reducing model size by 75% (1,390MB to 350MB on disk), offering a practical route to deployment in resource-constrained environments.

3 METHODS

3.1 Data Sources and Preparation

Table 7: Data modalities and their sources. For GMV GRIAL, training sample counts per motion class are: I=Inertial, SA=Safe, SP=Spin, SU=Sun, T=Tumbling, Y=Yaw.

Source	Notes	n	Labels
MMT-9	Real obs.	227,697	-
CASSANDRA	Simulated	800	160 anom., 640 normal
GMV GRIAL	Simulated	22,009	I:1259, SA:1940, SP:1280, SU:940, T:1333, Y:5131

The MMT-9 light curve dataset was downloaded as text files from <https://www.sao.ru/lynx/karpov/satellites/> and processed with custom scripts using the BeautifulSoup Python library (v4.11.1). The light curves varied in length

and were therefore resampled to $n_{\text{samples}} = 128$ time steps. Whilst the Perceiver architecture does not strictly require uniform input length, uniform resampling simplifies batching and ensures consistent positional encoding across samples. Visual inspection showed this length to not noticeably degrade light curves. Similarly, the anomaly detection (CASSANDRA) and motion prediction (GRIAL) fine-tuning datasets were processed to $(n_{\text{samples}}, 128, n_{\text{labels}})$, where there were two labels for anomaly detection and six for motion prediction. For details on dataset sizes, please see Table 7, which described the datasets produced by the CASSANDRA simulator (anomaly detection fine-tuning), and GRIAL simulator (motion prediction fine-tuning).

3.2 Model Architecture

Our Perceiver-VAE model architecture is outlined in Equation 1, and we briefly describe these steps below.

We implemented Perceiver as outlined in [8], adapting the Pytorch implementation of [24], including Fourier positional encoding (Figure 2, Equation 1). This encoding maps temporal coordinates to higher-dimensional representations, enabling the attention mechanisms to capture sequential patterns in light curve data.

To include Variational Autoencoder functionality we computed the mean and log variance of the learned latent vector z , using those values to sample the latent space probabilistically. This produces the reconstructed input, the Mean Squared Error loss (Equation 3, and KL Divergence (which constrains the latent space such that it is well ordered and stable, Equation 7). By including the VAE sampling, (as opposed to standard autoencoding), we create a continuous, structured latent space that enables generation of novel data points through interpolation and sampling. For decoding, we use a standard feed forward network with the architecture shown in (Equation 2).

$$\text{Perceiver-VAE} = \left\{ \begin{array}{l} \text{Encoder:} \\ \text{Input} \xrightarrow{\text{Fourier Encoding}} \text{Encoded Position} \\ \text{Latent Array} \xrightarrow{\text{Cross-Attention, FF}} \text{Updated Latents} \\ \text{Updated Latents} \xrightarrow{\text{Self-Attention}} \text{Processed Latents} \\ \text{Processed Latents} \rightarrow \mu, \log \sigma^2 \\ z = \mu + \varepsilon \cdot \exp(0.5 \cdot \log \sigma^2) \Big|_{\text{reparam.}} \\ \text{Decoder:} \\ z \rightarrow \text{Reconstruction} \end{array} \right. \quad (1)$$

Where

$$\text{Decoder} = \left\{ \begin{array}{l} \text{Linear}(d_{\text{latent}} \rightarrow 512) \\ \text{LayerNorm} \rightarrow \text{ReLU} \\ \text{Linear}(512 \rightarrow 1024) \\ \text{LayerNorm} \rightarrow \text{ReLU} \\ \text{Linear}(1024 \rightarrow d_{\text{output}}) \end{array} \right. \quad (2)$$

We evaluate three model scales: small (64 latents, $d_z=128$, 6.4M parameters), medium (128 latents, $d_z=128$, 13.2M parameters), and large (384 latents, $d_z=512$, 347.5M parameters), the latter representing the largest scale trainable within our computational budget. We note that magnitude pruning of the large model retains performance with 60–70% of weights removed,

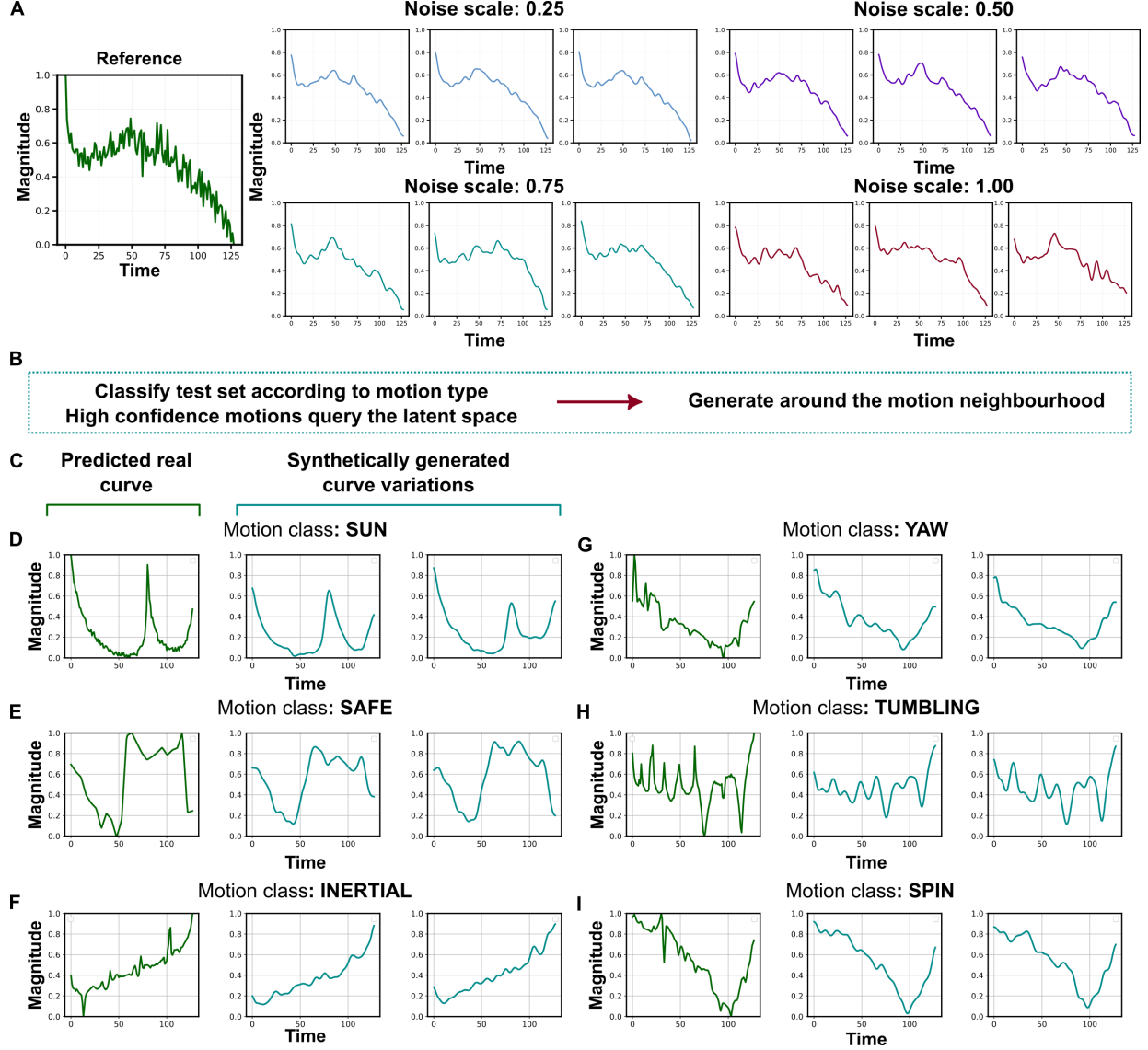


Figure 9: Sampling the latent space of our fine-tuned Perceiver-VAE. (A) Initial exploration of noise scaling in synthetic data generation. As in VAE training, at inference time we can offer a reference real light curve into the forward pass of the model. Once this has been encoded, we capture the latent vectors that respond highest to this reference curve, apply gaussian noise at different scales to these scalar values (0.25-1.0), and then decode the results. As expected, this results in synthetic light curves that adhere to the general global feature of the input curve, but differ on local features, and at increasing variance as you increase the distance from the latent activation. (B-C) Our synthetic data generation querying approach. Here we use the fine-tuned motion classifier, but this could also readily be the anomaly tuned classifier from Section 2.6. D-I To generate informed synthetic data, we first classify a held out test set using the motions described in Fig 8. Selecting a desired motion (e.g., Tumbling), we isolated test samples with high confidence class likelihood of belonging to tumbling, before using those as our latent space activating samples. As in (A), we then generate synthetic samples using a gaussian noise with noise scale of 0.75 around the latent activations.

suggesting substantial redundancy; we discuss this further in Section 4.

3.3 Pre-training strategy

We pre-trained our Perceiver-VAE model using a multi-task self-supervised learning approach. For this we incorporated three objectives: reconstruction, masking, and forecasting. The reconstruction task trained the model to encode and decode whole light curves, establishing the foundations for latent space

representation. At the same time, the masking task required the model to predict randomly masked segments of input sequences, encouraging our model to learn contextual relationships within the light curve. Finally, the forecasting task trained the model to predict future sequence values, promoting the capture of temporal dynamics and enabling forecasting at inference time.

Each sample is presented to the model once per batch in a single forward pass. The reconstruction, masking, and forecast-

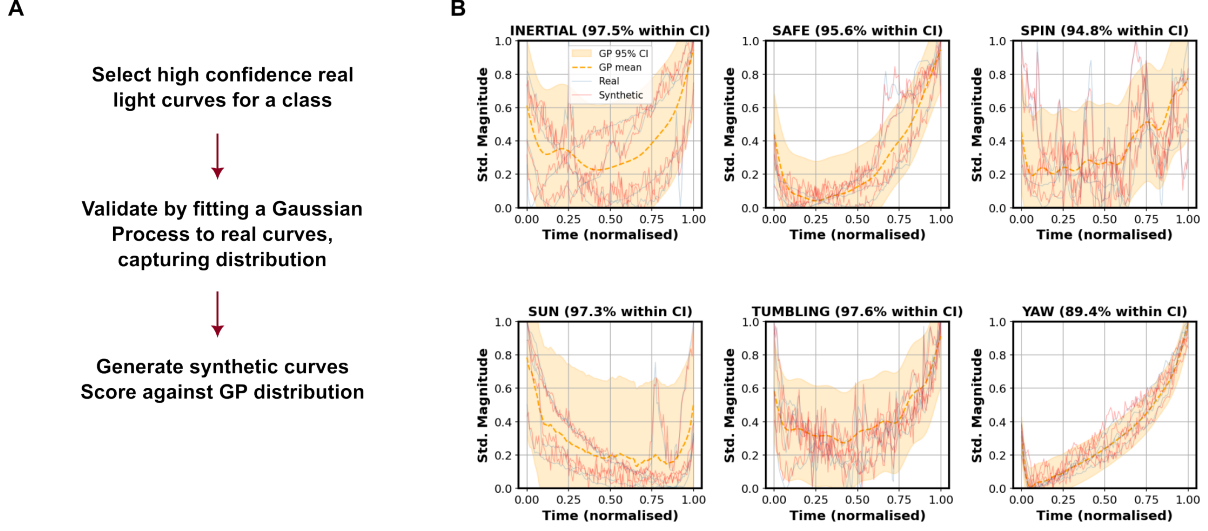


Figure 10: Gaussian Process validation of synthetic light curves. **(A)** Overview of the validation procedure: high-confidence real light curves are selected per class, a GP is fitted to the real distribution, and synthetic curves are scored against the resulting confidence band. **(B)** GP validation panels for all six motion classes. The orange band and dashed line show the GP 95% confidence interval and mean respectively, fitted to real curves (blue). Synthetic curves are shown in red. The score reported in each panel title denotes the fraction of synthetic curve time steps falling within the GP 95% confidence interval, averaged across all synthetic curves for that class. Five of six classes exceed 94%, with YAW showing a lower score (89.4%) consistent with the greater morphological variability introduced by grouping four sub-motions into a single class.

ing outputs are computed simultaneously within this pass, with each task operating on a different view of the same input (full sequence, randomly masked sequence, and future-masked sequence respectively). The KL divergence is therefore computed once per sample from the single latent sample, and the effective batch size equals the nominal batch size throughout pre-training. We combined these learning objectives by summing individual mean squared error loss components for each task, combined with a KL divergence term (weight $\alpha 0.001$) to regularise the latent space (Equation 3). This combined optimisation enabled the model to learn rich representations of light curve characteristics while maintaining a structured latent space suitable for generation and interpolation.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \alpha \mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{mask}} + \mathcal{L}_{\text{forecast}}, \quad (3)$$

where:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2, \quad (4)$$

$$\mathcal{L}_{\text{mask}} = \frac{1}{N_m} \sum_{i \in M} (x_i - \hat{x}_i^m)^2, \quad (5)$$

$$\mathcal{L}_{\text{forecast}} = \frac{1}{N_f} \sum_{i \in F} (x_i - \hat{x}_i^f)^2. \quad (6)$$

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2). \quad (7)$$

The total loss (Equation 3) combines four components: The reconstruction loss ($\mathcal{L}_{\text{recon}}$, Equation 4) measures the Mean Square Error (MSE) between the original inputs x_i and reconstructed outputs $\hat{x}_i$ across all N data points.

The masking loss ($\mathcal{L}_{\text{mask}}$, Equation 5) calculates MSE on a subset of masked inputs (M), where $\hat{x}_i^m$ represents reconstructions of deliberately masked elements, encouraging the model to infer missing data from context.

The forecasting loss ($\mathcal{L}_{\text{forecast}}$, Equation 6) computes MSE on future time steps (F), where $\hat{x}_i^f$ represents predicted values for future time points, training the model to extrapolate temporal patterns.

The KL divergence loss ($\mathcal{L}_{\text{KL}}$, Equation 7) regularises the latent space by ensuring the learned distribution (parameterised by mean μ_j and variance σ_j^2 across d latent dimensions) approximates a standard normal distribution, weighted by hyperparameter α .

Pre-training hyperparameters were selected via Bayesian optimisation using Optuna [25], running 25 trials of 15 epochs each per model scale. The optimal learning rates were 1.41×10^{-3} , 5.56×10^{-3} , and 5.38×10^{-4} for small, medium, and large respectively, with batch sizes of 16, 32, and 16. All models were trained with gradient clipping at 0.5 [26] to prevent exploding gradients, which we observed empirically during the early training phase where the KL divergence term increases rapidly before stabilising (Figure 3). A learning rate scheduler reducing the learning rate by a factor of 0.5 when validation loss plateaued for 5 consecutive epochs, and early stopping with a patience of 50 epochs.

All pre-training was conducted on the baskerville HPC cluster, on single a40 and a80 GPUs.

3.4 Fine-tuning

To fine-tune our pre-trained Perceiver-VAE, we freeze the encoder weights and train a lightweight classification head on top of the learned latent representations [27]. For a given input light curve, the frozen encoder produces a mean latent vector μ , which is passed directly to the classification head. The optimisation objective is therefore simply the categorical cross-entropy loss:

$$\mathcal{L}_{\text{fine-tune}} = \mathcal{L}_{\text{class}} = -\frac{1}{m} \sum_{i=1}^m \sum_{c=1}^C y_{ic} \log(\hat{y}_{ic}), \quad (8)$$

across m samples and C classes, where y_{ic} are the true labels and $\hat{y}_{ic}$ the predicted probabilities. This approach follows standard practice for fine-tuning foundation models, using the pre-trained encoder purely as a feature extractor.

The classification head is a two-layer MLP: a linear projection to a hidden dimension, followed by ReLU activation, dropout regularisation ($p = 0.3$), and a final linear projection to the number of output classes.

We trained using the Adam optimiser with a learning rate of 1×10^{-3} for all fine-tuning conditions. Batch size and maximum epochs were set per task: anomaly detection used a batch size of 32 for up to 50 epochs, and motion prediction used a batch size of 64 for up to 150 epochs, reflecting the difference in dataset size between the two tasks. To address class imbalance in the motion prediction task (Table 6), we applied inverse-frequency class weighting to the cross-entropy loss. The best checkpoint was selected by validation ROC AUC.

To evaluate the contribution of pre-training, we additionally train classification heads on randomly initialised Perceiver encoders of equivalent scale, with weights drawn from the default PyTorch initialisation and held frozen throughout fine-tuning. These random encoder (r.e.) variants are otherwise identical to the pre-trained models, isolating the effect of pre-training on downstream performance.

3.4.1 Anomaly Detection Fine-tuning

For anomaly detection, the classification head produces a binary output distinguishing anomalous from normal light curves. We evaluated on a stratified held-out test set comprising 20% of the CASSANDRA dataset, reporting both accuracy and ROC AUC.

To evaluate label efficiency, we additionally conducted a k -shot study comparing the pre-trained Perceiver against the CNN baseline. For each value of $k \in \{2, 5, 10, 25\}$, k labelled examples per class were sampled from the training pool using independent random seeds across 20 independent runs. Each run trained the classification head (or CNN) for 50 epochs using the Adam optimiser with a learning rate of 1×10^{-3} and batch size of 32, and was evaluated on the same stratified held-out test set used in the main experiments.

3.4.2 Motion Prediction Fine-tuning

For motion prediction, the classification head produces a six-class output corresponding to the grouped motion classes described in Table 6. We similarly evaluated on a stratified 20%

held-out test set, reporting per-class precision, recall, F1-score, and ROC AUC.

3.5 Synthetic Data Generation

To explore the latent space and synthetic data generation capabilities of our model, we implemented a neighbourhood sampling approach. For synthetic data generation, our objective is to sample the latent space in a way that produces meaningful variations of a reference curve while preserving its class identity.

Here we use the fine-tuned motion prediction classifier to generate a set of high confidence ‘query’ curves. The algorithm first identifies some reference curves above a given confidence threshold (0.9) for a requested motion class e.g., tumbling. Then, it uses these to index the learned latent space of the model, and samples around that latent space by adding Gaussian noise to the highest activating neurons. For visual clarity in Figure 9, Gaussian smoothing ($\sigma = 2.0$) is applied to the displayed synthetic curves to aid inspection of global morphological features. The GP validation (Section 2.7, Figure 10) is performed on unsmoothed outputs, confirming that quantitative synthetic curve quality is maintained without smoothing.

To quantitatively validate generated synthetic light curves, we fitted a Gaussian Process (GP) to the real light curve distribution for each motion class and evaluated whether synthetic curves fell within the learned distributional bounds.

For each class, high-confidence real curves (predicted class probability > 0.75 from the fine-tuned motion classifier) were selected, and a GP was fitted to these using an RBF kernel with additive white noise: $k(t, t') = \text{RBF}(l=0.2) + \text{WhiteKernel}(\sigma^2=0.01)$, with kernel hyperparameters refined via three restarts of marginal likelihood optimisation and outputs normalised prior to fitting. This provides a non-parametric estimate of the mean and standard deviation of the real light curve distribution at each of the 128 resampled time steps.

Each synthetic curve was scored by computing the fraction of time steps for which the absolute deviation from the GP posterior mean fell within two standard deviations:

$$s = \frac{1}{128} \sum_{i=1}^{128} \mathbf{1}[|\hat{y}_i - \mu_i| < 2\sigma_i], \quad (9)$$

where μ_i and σ_i are the GP posterior mean and standard deviation at time step i , and $\hat{y}_i$ is the synthetic curve value. The class-level score is the mean of s across all synthetic curves for that class. This treats the GP as a distributional reference for plausible real curves, such that a high score indicates synthetic curves are statistically consistent with real observations under the GP model.

3.6 Model Compression

To assess parameter efficiency of the large Perceiver-VAE, we applied global unstructured magnitude pruning [28] separately to the Perceiver layers and the VAE bottleneck (mean and log-variance projections), sweeping sparsity from 10% to 90%. Following pruning, batch normalisation statistics were recal-

brated on 500 training samples prior to evaluation on the held-out test set. Post-training dynamic quantisation was applied to the full model, converting all linear layers to INT8 precision using PyTorch’s `quantize_dynamic`. No calibration data was required.

4 OUTLOOK

Our work is the first to demonstrate that a VAE architecture can effectively learn meaningful representations from light curve data and be fine-tuned for multiple downstream tasks including detecting anomalous space object behaviour and motion prediction. We have also shown that our approach scales well to large datasets while maintaining computational efficiency, which is critical considering the rapidly growing amount of Space Object data. Here, we developed a two-stage approach, first through unsupervised pre-training, whereby the model learns rich features which aid reconstruction of real light curves. Then we performed supervised fine-tuning, which leverages those features for both anomaly classification, reaching 85% accuracy with a 0.92 ROC AUC score, and attitude mode motion classification, achieving 82.7% test accuracy, with 0.95 ROC AUC scores. Additionally, we demonstrate that the same architectural framework and approach can simultaneously achieve multiple space object behavioural analysis goals (i.e., anomaly detection, attitude mode classification, and synthetic data generation), provided that the representations learned during pre-training are diverse and rich, and that these are coupled with high-fidelity fine-tuning simulation data. Although in this study we identify several predicted anomalous light curve modes, some present clearly anomalous behaviour (e.g., rapid tumbling), while for others the underlying behaviour is less clear. A further direction is per-object anomaly characterisation: for well-represented objects in the pre-training data, fitting a Gaussian Process to the distribution of typical light curves for a specific object and comparing high reconstruction error curves against this baseline would allow more rigorous validation of whether these curves represent genuinely atypical behaviour for that object specifically. Systematic cataloguing and analysis of simulation conditions that lead to these various curve morphologies would allow us to verify whether particular erroneous space object dynamics produce these curves (e.g., in Fig 7C, F). A further valuable direction is a systematic pre-training scaling study, training models on increasingly large subsets of the MMT-9 dataset to characterise how pre-training data volume impacts performance. This is of particular relevance to the SDA context, where pre-training data may grow substantially as sensor networks expand, but detailed labels remain sparse.

From this work, several key directions for further development emerge. Firstly, development of robust quantitative benchmarks for AI synthetic data. Whilst our approach encourages plausibility in the generated curves through reference-based sampling, systematic evaluation frameworks are needed to validate produced synthetic data beyond visual inspection/sense checks. Whilst we provide GP-based validation of synthetic light curve quality in this work, further quantitative evaluation metrics such as KL divergence, power spectral density comparisons, or Earth mover distance between real and synthetic distributions remain valuable directions for future work.

For example, these frameworks could assess statistical distribution matching between real and AI generated datasets, and evaluate like-for-like comparisons in downstream tasks trained on real data, numerically simulated data, and/or data generated through our approach (i.e., generative AI data). Finally, physics-informed neural networks (PINNs) could be used to quantify physical plausibility. Briefly, these networks combine a data loss term with a physics loss term. For example, for light curves this might be a reflectivity model as outlined in [22]. At inference time, a well-trained PINN will give a good indicator as to whether a synthetically generated light curve is obeying the underlying physics due to the value of the physics loss term.

A key limitation of the current work is the simulation-to-real gap: both fine-tuning datasets (CASSANDRA for anomaly detection and GRIAL for motion prediction) are simulated, whilst pre-training data and any operational inference data are observations. Whilst the pre-trained model demonstrates some unsupervised anomaly discrimination on the CASSANDRA dataset (ROC AUC 0.71 without any labels), the domain gap between real and simulated curves limits threshold-based detection, motivating fine-tuning as a necessary bridging step. Closing this gap will require either real labelled anomaly data, which is currently scarce due to national security and commercial sensitivities, or domain adaptation techniques that better align simulated and real light curve distributions. Contrastive approaches using paired real and simulated observations of the same object could be a promising direction, as could physics-informed data augmentation to make simulated curves more representative of real observational conditions.

Finally, magnitude pruning experiments (Section 2.8) suggest the large model contains substantial redundancy in its Perceiver layers, motivating future work on more parameter-efficient pre-training strategies. Similarly, the negligible accuracy cost of INT8 quantisation suggests that deployment of large-scale space object characterisation models in resource-constrained operational environments is feasible.

Another key direction is privacy and security considerations. As generative AI (and therefore also synthetic data) becomes more prevalent in SDA, privacy/security need to be considered to ensure that if a model is pre-trained on sensitive light curves that they cannot be extracted from the resulting fine-tuned model at inference time. Additionally, the quality of a self-supervised model like ours is dependent on the quality of both the pre-training and fine-tuning data. If models such as these are to be integrated into critical Space Domain/Situational Awareness pipelines they become an attractive target for adversarial attack, particularly through data poisoning. For this reason, ensuring the integrity of data provenance is highly important, and blockchain approaches could be used here, treating the dataset as the transaction to be verified on the public ledgers, e.g., through the use of Trustchain [29].

A substantial challenge direction for further development is multi-modal fusion - i.e., integrating the light curve analysis presented here with other observation types (e.g., hyperspectral data, orbits, radar etc.) Indeed, part of the decision to use a Perceiver based architecture was to readily enable extension to other data modalities, as it is well suited to this through its

cross-attention and latent bottleneck design, which could enable multiple modalities to be reduced to a common set of embeddings. A natural extension of future work is to exploit the Perceiver's ability to handle variable-length inputs directly, removing the uniform resampling constraint applied here and allowing the model to process light curves at their native sampling rates. However, there are substantial technical challenges in multimodal fusion, including developing an appropriate fusion strategy at different levels of the pre-training/fine-tuning pipeline. One such method to fuse information from other sensors with light curves could include implementation of contrastive learning using paired observations. For example, positive pairs (same space object at different dates) and negative pairs (different objects on the same date) would help develop more discriminative features for space object characterisation. Additionally incorporating metadata prediction tasks as part of the self-supervised framework could help capture characteristic signatures of specific satellites, or debris. For example, a fine-tuned metadata prediction FM could generate a likely light curve given an orbital eccentricity, which could help unpick the relationships between satellite structure/behaviour and orbital path.

Our initial framework provides promising results, and as the amount of data available increases, a basis for further analysis. We suggest Foundation Models, in particular those that exploit multimodal datasets, could become powerful tools for automated space object monitoring and space situational awareness. As orbital populations continue to grow, automated approaches such as ours will become increasingly crucial for maintaining space safety & sustainability.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

AUTHOR CONTRIBUTIONS

CRedit Classification:

Conceptualization: IG, MV, VN, **Data Curation:** IG, AC, JF, **DRR Formal Analysis:** IG, **Funding Acquisition:** PM, MV, VN, **Investigation:** IG, **Methodology:** IG, MV, VN, **Project Administration:** IG, MV, VN, **Software:** IG, AC, **Supervision:** VN, **Writing – Original Draft Preparation:** IG, **Writing – Review & Editing:** IG, VN.

5 DATA AVAILABILITY STATEMENT

Research data are not shared.

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SUPPLEMENTARY: BASELINE MODEL ARCHITECTURES

All baseline models take a light curve of length 128 as input and are trained from scratch using the Adam optimiser with a learning rate of 1×10^{-3} , batch size of 32, and dropout of $p = 0.3$ throughout.

RNN: A two-layer recurrent neural network with tanh activation and hidden size 64, following a standard stacked configuration to capture both short and long-range sequential dependencies. The final hidden state is passed to a linear output projection.

LSTM: A two-layer Long Short-Term Memory network with hidden size 64, again using a stacked configuration for multi-scale temporal modelling. The final hidden state is passed to a linear output projection.

CNN: Three successive 1D convolutional blocks with progressively increasing channel depths (32, 64, 128) and kernel sizes of 7, 5, and 3 respectively, to capture hierarchical temporal features at decreasing scales. Each block is followed by ReLU

activation and max pooling. Global average pooling precedes a linear output projection.

Transformer: A two-layer Transformer encoder with model dimension $d_{\text{model}}=64$, 4 attention heads, and feedforward dimension 256, following the standard 1:4 model-to-feedforward scaling convention [30]. Input timesteps are projected to d_{model} before encoding, and global average pooling over the sequence precedes a linear output projection.